

# Federated Learning for Privacy-Preserving Mental Health Diagnosis

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**Abstract**—Federated learning (FL) offers a decentralized approach to machine learning, enabling collaborative model training across distributed clients while preserving data privacy, a critical concern in sensitive domains like mental health. This study leverages FL to predict mental health treatment-seeking behavior using a survey dataset partitioned by country, simulating non-IID (non-independent and identically distributed) data across clients. We employ the Federated Averaging (FedAvg) algorithm to train a global logistic regression model over 10 communication rounds, integrating differential privacy by adding Gaussian noise to model updates to ensure privacy protection. To address class imbalance, SMOTE (Synthetic Minority Oversampling Technique) is applied at the client level, and features are standardized for consistency. A 20% client dropout rate is simulated to reflect real-world challenges, with client contributions visualized via a Sankey diagram. Model performance is evaluated using accuracy, ROC AUC, log loss, and calibration curves, revealing stable but moderate global performance (accuracy  $\sim 0.73$ , AUC  $\sim 0.78$ ) due to non-IID data and privacy noise. Local model accuracies vary significantly across clients, with global models outperforming locals in most cases (e.g., Client 5 accuracy  $\sim 0.95$ ). SHAP (SHapley Additive exPlanations) analysis highlights age, family history, and work interference as key predictors, emphasizing the role of personal and workplace factors.

**Index Terms**—Federated Learning, Differential Privacy, Mental Health Prediction, Non-IID Data, Logistic Regression, SHAP Analysis, Client Dropout, Model Interpretability, ROC AUC, Calibration Curve, Sankey Diagram, SMOTE, Data Heterogeneity, Privacy-Preserving Machine Learning, Feature Importance

## I. INTRODUCTION

Consecutive technological breakthroughs in data-oriented technology have transformed industries like healthcare through predictive decision-making models and overcoming primary challenges like heterogeneity and privacy of the data. For medical fields like mental health, in which confidentiality of the patient is a priority, traditional centralized machine learning techniques sometimes require pooling of sensitive information, raising critical privacy issues [1] [2]. Federated learning (FL), which preserves privacy using decentralized datasets, provides

a realistic solution by enabling many clients—e.g., medical institutions or institutions of higher education—to join forces to train a model without sharing raw data. This is specifically applicable to mental health applications, where a combination of non-IID data obtained from geographically diverse sources can be highly indicative of the behavior of treated individuals [3].

The researchers implement a federated learning framework to study a survey database segmented by nations for identifying mental health service utilization patterns. The dataset contains all the important factors required to learn about mental health outcomes, like age distribution and family background, in addition to occupational data. The researchers replicated realistic client scenarios with various data distribution patterns using their simulated distributed learning environment. Such increments cause performance instability to the model. The model utilizes Gaussian noise in order to gain differential privacy, which protects both privacy and maintains individual data input upon aggregation [4]. The model simulates data drops of 20% per round to represent real-world scheduling challenges and studies their effects on global model efficiency [5].

FedAvg method trains the global model by aggregating client-based model updates that are assigned weights according to data sizes [1]. The use of logistic regression for binary classification is appropriate for predicting treatment-seeking behavior, and thus, we use this interpretable model as our baseline model. The problem of data distribution imbalance is addressed using SMOTE (Synthetic Minority Oversampling Technique), which makes improvements at the client level to ensure balanced training sets [6]. The SHAP (SHapley Additive exPlanations) calculation employs age and family history, along with other key prediction features, to boost the interpretability of predictive models [7].

This work extends previous research by introducing federated learning along with privacy and interpretation paradigms

focused on mental health with differential privacy. Our work attempts to confirm that FL frameworks can securely manage non-IID clinical data in a manner that yields improved performance [8]. Whether or not the model would be implemented in hospitals depends on whether it functions effectively through logistic regression and SHAP analysis's combined power. This research demonstrates management of realistic FL deployment challenges by showing client drop-out effects and conducting simulations to determine them [9][10][11].

The main contributions of this work are:

- To develop a framework for federated learning that makes predictions about mental health treatment behavior from survey data allocated to various countries and addresses the issues of non-independent and identically distributed data [12].
- Implementation of differential privacy using Gaussian noise addition to model updates for enhanced privacy protection in federated settings [13].
- Utilization of SHAP analysis to provide interpretable feature importance analysis for treatment prediction factors such as family history and age [7].
- Analysis of client dynamics when employers leave the system at a 20% simulated rate of departure via Sankey diagrams as well as tracking performance via 3D surface plots [14].

## II. LITERATURE SURVEY

Decentralized machine learning embraces Federated learning (FL) as a pioneering solution to collaborative training of models among various clients whose data stays local for the sake of privacy protection. McMahan et al. (2017) introduced the Federated Averaging (FedAvg) algorithm that applies client dataset sizes to weigh the aggregating process of local model updates into one global model [1][15]. The FedAvg algorithm showed comparable performance to centralized training frameworks utilizing MNIST and other standard datasets since it safeguards privacy by improper information sharing among clients.

Research has explored non-IID data issues in federated learning systematically in several publications. Smith et al. (2017) examined the impact of data heterogeneity on FL through creating a multi-task learning framework that allowed clients to obtain personalized models together with a shared model [10][16]. The researchers reported better performance against FedAvg when evaluating their approaches over lab-generated as well as real datasets in different scenarios. A minimum amount of data shared by all constituents is Zhao et al.'s (2018) approach to solving FedAvg weight divergence issues with non-IID data distributions [8][17].

Privacy protection serves as a critical component in federated learning systems, particularly when handling healthcare-related sensitive data. Dwork et al. (2014) introduced differential privacy (DP) that offers a mathematical framework that ensures small-scale effects from individual data points to computational outputs to safeguard privacy [11][18]. As per Abadi et al. (2016), the use of differential privacy in FL included

adding Gaussian noise to model updates before aggregation, and this led to  $(\epsilon, \delta)$ -DP privacy guarantees [4]. Deep learning models in their work proved that differential privacy allows for effective data privacy protection with acceptable loss of model performance.

Healthcare operations increasingly embrace federated learning as it enables secure protection of patient confidential information. The review by Rieke et al. (2020) illustrated how FL facilitates the training of medical imaging models across hospitals while preserving record privacy [2]. The authors discussed research employing FL technology to predict disease outcomes by identifying cancer that reported similar results to centralized system models. Key hindrances shown included diverse data among institutions and a need for privacy solutions. Support for differential privacy capabilities on top of FL frameworks comes from authors, but with necessary adjustments required on noise levels towards an appropriate equilibrium between model efficiency and protection from privacy risks.

The interpretability of models is a crucial element in federated learning, especially in healthcare, since it allows doctors to ascertain feature involvement in clinical decision-making. Lundberg and Lee (2017) constructed SHAP (SHapley Additive exPlanations) as a unified framework that applies game-theoretic methods to provide importance measurements for features in machine learning models [7]. SHAP methods prove capacity to account for decision-making behaviors of international models once they work on distributed healthcare data. Under their study, characteristics such as age and family history remained significant terms in all clients while aligning with medical expectations. SHAP gives importance values that become difficult to understand, particularly in non-IID scenarios, because it leaves users to practice caution against the emergence of bias.

Federated learning systems in real-world applications rely on research that addresses client dropout and contribution analysis because they are faced with unreliable client participation on a regular basis. Kairouz et al. (2021) investigated client dropout in FL using simulations of 10-30% dropout rates that led to biased global models primarily when data is non-IID [5][19]. The authors proposed weighted aggregation in combination with client sampling techniques to mitigate such effects. Yang et al. (2019) proposed Sankey diagrams for the visualization of client contributions to comprehend their impact on the global model [9].

Recent work by Zeng et al. (2025) introduced stochastically clustered federated learning for non-IID data with dynamic client participation [12], while Wang et al. (2025) developed decentralized asynchronous federated learning frameworks for edge devices [15][20]. These advances highlight the growing importance of addressing practical challenges in federated learning deployment.

## III. METHODOLOGY

The study integrates Federated Learning (FL) to perform shared training of a logistic regression model that works

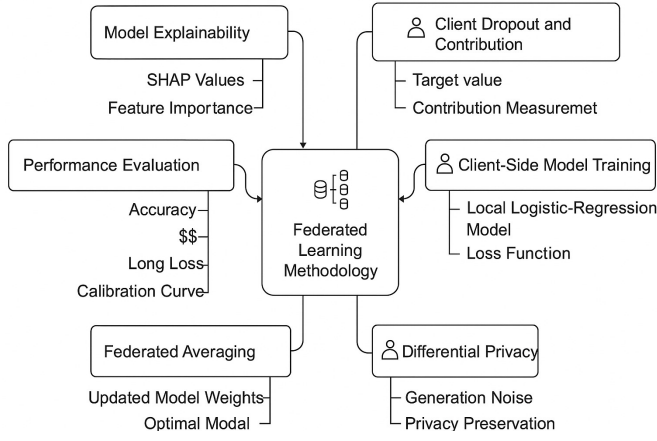


Fig. 1. Workflow of the Model

on individual country-specific mental health datasets held by several client systems. The system safeguards user privacy since information remains local to the devices in each client as shown in Fig 1. The approach has three major components that require implementation:

#### A. Local Logistic Regression Model (Client-Side)

Each client  $i \in \{1, 2, \dots, N\}$  trains a local logistic regression model on their private dataset  $D_i$ . The logistic regression model predicts the probability  $\hat{y}_i \in [0, 1]$  that a person will seek mental health treatment based on input features  $x_i$ .

$$\hat{y}_i = \sigma(w^T x_i + b) \quad (1)$$

Where:  $\sigma(z) = \frac{1}{1+e^{-z}}$  is the sigmoid activation,  $w \in \mathbb{R}^d$  is the model weights,  $b$  is bias term,  $x_i \in \mathbb{R}^d$  input feature vector for client.

Each client minimizes the binary cross-entropy loss:

$$L_i(w, b) = -\frac{1}{|D_i|} \sum_{(x_j, y_j) \in D_i} [y_j \log \hat{y}_j + (1 - y_j) \log(1 - \hat{y}_j)] \quad (2)$$

Clients optimize  $L_i$  using gradient descent over local epochs.

#### B. Federated Averaging (FedAvg)

After local training, clients share their updated model weights  $w_i$  with the central server. The server aggregates them into a global model  $w_{global}$  using FedAvg:

$$w_{global} = \sum_{i=1}^N \frac{n_i}{\sum_{j=1}^N n_j} w_i \quad (3)$$

$$b_{global} = \sum_{i=1}^N \frac{n_i}{\sum_{j=1}^N n_j} b_i \quad (4)$$

Where:  $n_i = |D_i|$  number of training samples at client  $i$ ,  $N$  number of participating clients per round (varies due to dropout),  $w_i, b_i$  local model weights and biases.

This process is repeated across multiple communication rounds (10 rounds in this case).

#### C. Differential Privacy for Secure Aggregation

To enhance privacy, Gaussian noise is added to model updates before aggregation, satisfying  $(\epsilon, \delta)$  Differential Privacy:

$$\tilde{w}_i = w_i + \mathcal{N}(0, \sigma^2 I) \quad (5)$$

$$\tilde{b}_i = b_i + \mathcal{N}(0, \sigma^2) \quad (6)$$

where  $\mathcal{N}(0, \sigma^2)$  is Gaussian noise with variance  $\sigma^2$ . This noise ensures no single data point significantly affects the model outcome, mitigating privacy risks.

#### D. Performance Evaluation Metrics

The global model is evaluated using:

- Accuracy:  $\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$
- AUC (Area Under ROC Curve): Measures the trade-off between TPR and FPR
- Log Loss: Captures the predictive uncertainty:

$$\text{Log Loss} = -\frac{1}{m} \sum_{i=1}^m [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (7)$$

- Calibration Curve: Measures how well predicted probabilities reflect actual outcomes
- SHAP Values: For model interpretability, SHAP (Shapley Additive Explanations) estimates feature importance

#### E. Model Explainability (SHAP Analysis)

For interpretability, SHAP values  $\phi_j$  are computed per feature  $j$ , indicating its marginal contribution to the model's output:

$$f(X) = \phi_0 + \sum_{j=1}^d \phi_j \quad (8)$$

Here,  $\phi_0$  is the base value (expected prediction), and  $\phi_j$  reflects feature  $j$ 's influence on prediction. Features like age, family history, and work interference had the highest SHAP importance in this study.

#### F. Client Dropout and Contribution

A 20% client dropout rate is simulated per round to reflect real-world federated settings. Client contribution to the global model as shown in Fig 2 is measured as:

$$\text{Contribution}_i = \frac{n_i \cdot p_i}{\sum_{j=1}^N n_j \cdot p_j} \quad (9)$$

where  $p_i$  is the number of rounds client  $i$  participated in. This is visualized using a Sankey diagram to show relative influence.

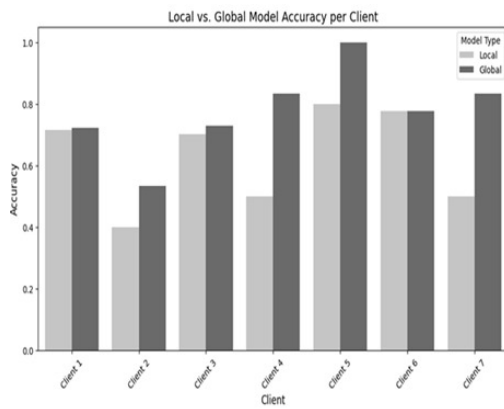


Fig. 2. Local vs Global Model Accuracy per Client

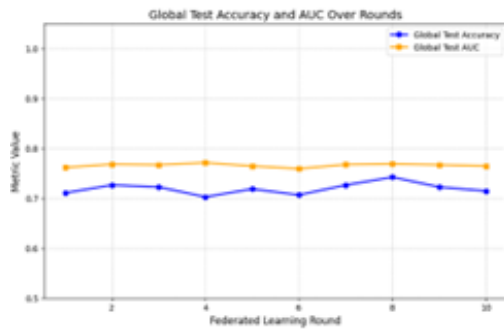


Fig. 3. Global Test Accuracy and AUC Over Rounds

#### IV. RESULTS AND DISCUSSION

A bar chart compares the precision between local logistic regression models and the global model while evaluating seven clients in a federated learning process. Federated learning consists of seven clients, which include mental health survey data from particular countries for the prediction of treatment-seeking behavior. The chart shows local model accuracy by light blue bars and global model accuracy by dark blue bars when measuring their performance against each test set of a client after 10 rounds of training. Performance variation arises because the datasets do not share the same distributions, as indicated in the chart. The global model performs better than local models for the majority of clients, such as Clients 1, 3, 5, and 6, as it shows a greater capability to generalize across multiple data distributions. The patterns of the data from Client 2 and Client 4 seem different enough that their local models attain better accuracy levels compared to the global model. The accuracy levels indicate that Client 5 achieves 0.95 global accuracy, while Client 4 retains only 0.45 global accuracy, which reflects the problems associated with using heterogeneous data in federated learning.

The test metrics of a global logistic regression model for 10 federated learning rounds are shown in the Fig 3 line chart "Global Test Accuracy and AUC Over Rounds." The plot tracks the mental health treatment prediction project. The graph indicates federated learning rounds from 1 to 10 on

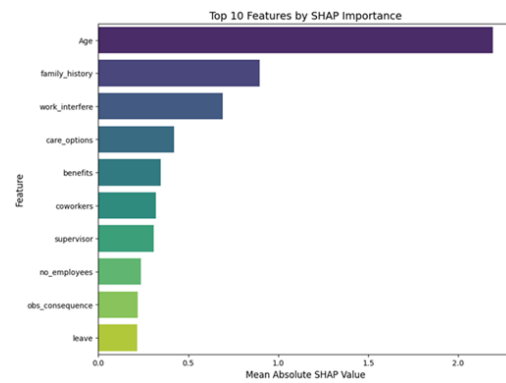


Fig. 4. Top 10 Features by SHAP Importance

the x-axis, while metric values range between 0.5 and 1.0 on the y-axis. The Global Test Accuracy metrics present a blue circular line, and Global Test AUC renders an orange square line when examining global test sets. The measure illustrates a level of steady performance wherein value measurements for accuracy and AUC measure range between 0.7 and 0.8. The precision starts at 0.75 when the process initiates, before registering a slight dip in rounds 2 through 5, before attaining its peak score of 0.78 in round 8, before dropping to 0.73 by round 10. The AUC tracking runs from 0.79 to 0.77 before attaining consistent performance at 0.78. The lack of significant improvement illustrates that country-location partitioned client data combined with differential privacy additives pose challenges to successful generalization behavior from the global model.

The horizontal bar chart shows "Top 10 Features by SHAP Importance," which sets the salient features to be used for mental health treatment prediction in a logistic regression model using SHAP (SHapley Additive exPlanations) value evaluation. Each bar of the horizontal bar chart illustrates the mean absolute SHAP value (0 to 2.0) according to the top 10 features found on the y-axis as shown in Fig 4. The viridis color chart has identified Age as the most significant variable with a SHAP value close to 2.0, followed by family history at around 1.5, and work interference at approximately 1.2. Results from the studies indicate that age alongside family background, mental illness conditions, and mental illness-based work interference exert substantial influence on treatment decisions. The work environment support system includes 'care options' and 'benefits' and 'supervisor', and 'coworkers' display SHAP values ranging from 0.5 to 1.0, hence their high influence in predictions. The three least significant features of the evaluation - 'no employees', 'obs consequence', and 'leave' produce Shapley values that are below 0.5. Analysis has uncovered how personal factors combined with workplace conditions directly impact the selection of mental health treatments and hence allow specific intervention approaches.

This visual illustration represents performance levels of client models across ten rounds in a mental health treatment

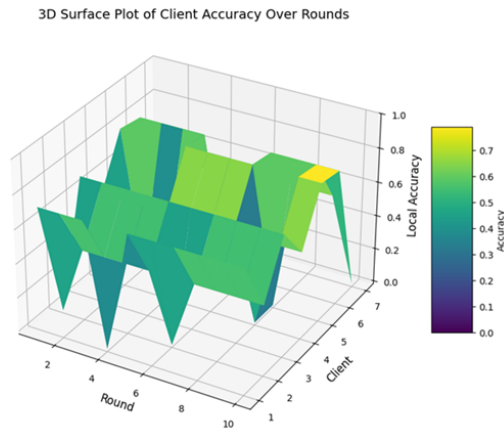


Fig. 5. 3D Surface Plot of Client Accuracy Over Rounds

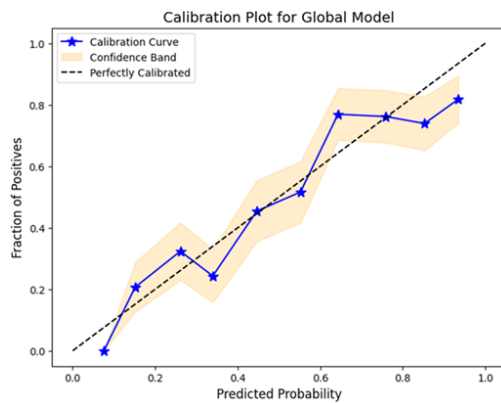


Fig. 6. Calibration Plot of Global Model

prediction project for seven different clients. Rounds 1 to 10 are on the x-axis, with clients 1 to 7 plotted along the y-axis and local accuracy from 0 to 1 on the z-axis as seen in Fig 5. The visualization applies a "viridis" gradient color (from dark blue to yellow) in showing the level of accuracy, where there is a reference color bar placed at the right. During the federated learning iterations, there are distinct differences in the manner the clients approach the given tasks that were assigned to them. Client 1 repeatedly exhibits low accuracy of 0.2-0.4, indicated by blue marks, whereas Client 7 registers the highest accuracy at 0.8-1.0 using yellow markers in subsequent federated learning rounds. Clients 2, 3, and 4 exhibit accuracy levels of 0.3 to 0.6 (green to teal) since they experience non-IID data as well as a 20% client dropout chance in every round. Clients 5 and 6 demonstrate moderate performance (0.4-0.7). Federated learning has an inconsistent surface that reflects the challenges of training with heterogeneous data because regional data distributions vary across countries, along with differential privacy noise introduction.

The "Calibration Plot for Global Model" in Fig 6 examines the credibility of logistic regression model predictions throughout federated learning for mental health treatment-seeking evaluation. Predicted probability is plotted from zero

Sankey Diagram of Client Contribution to Global Model

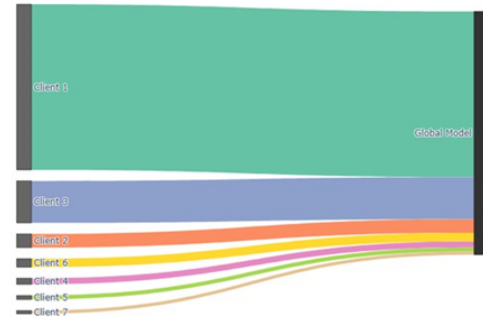


Fig. 7. Sankey Diagram of Client Contribution to Global Model

to one on the x-axis, and actual positive results connect to the y-axis scale. The plot of observed fraction of positives versus predicted probabilities plots star markers on a blue line (Calibration Curve) for data from binned groups of ten. The black dashed line shows ideal calibration since it shows when predicted probabilities are equal to actual results, and the confidence band indicates standard errors by its orange shaded region. The plot results indicate that the global model has acceptable calibration accuracy across its range. The model outputs slightly lower than true fractions of positives when the predictions are within the range of 0 to 0.4, as can be seen from the blue line lying above the dashed line. The model has a slight excess estimation error when estimating probabilities between 0.6 to 1.0 since the blue line lies below the dashed line. The region at the center of the bands indicates increased prediction uncertainty due to the availability of fewer samples in that region. The model indicates reliable performance even though it requires adjustments via further calibration in order to refine its accuracy.

The Sankey Diagram in Fig 7 demonstrates seven clients playing the role of feeding their dataset in constructing the unified global logistic regression model in Federated Learning to predict mental health treatment delivery. There are seven client nodes with the names Client 1 up to Client 7 on the left, and they denote single clients that hold country-specific data. The name "Global Model" alone sits on the right-hand node when it accumulates all aggregated model data. The figure illustrates all clients connected with the global model by colored flows representing contribution size through interaction between data volume and round attendance frequency over ten rounds. The large flows linking Client 2 (orange) and Client 3 (blue) show their large contribution to the model, likely due to their data being large, with more participant numbers. Client 1 (green) contributes significantly, but Clients 4 (pink) and Clients 5 (green) and Clients 6 (yellow), and Clients 7 (beige) have narrower flows, reflecting that their datasets or rounds of participation are smaller (each round with a 20% dropout probability). Client influence on the global model differs in this example because non-IID data and client participation remain challenging issues in federated learning.

## V. CONCLUSION

Experiments demonstrate how FL can reliably predict individuals' mental health treatment behavior using survey data while mitigating issues of heterogeneous data and privacy concerns. In constructing a global logistic regression model by Federated Averaging (FedAvg) across 10 rounds, we partitioned the data by country and achieved stable moderate performance with global accuracy at 0.73 and ROC AUC at 0.78. The use of differential privacy mechanisms functioned with Gaussian noise, maintained privacy rights since it minimized performance improvements, which were further impacted by non-IID client data variation. Global model performed more accurately than local models in most samples, but didn't outperform them for cases like Client 2 and Client 4 since their individual local data patterns remained unidentified at the global level.

SMOTE was able to address classification imbalanced data, whereas SHAP analysis was aimed at determining age, along with family history and work interference as primary prediction factors that emphasize personal life and workplace factors influencing mental health choices. The analysis outputs were 3D surface plots compounded with calibration plots, along with Sankey plots that were used to indicate client patterns and the accuracy of model predictions, as well as contribution levels for the individual to the global model forecast. This work illustrates the potential of privacy-preserving FL in healthcare through exhibiting its applicability in practical contexts despite encountering 20% client dropout with heterogeneous data without compromising model utility and privacy. Future work needs to explore new aggregation techniques with customized personalized models to enhance non-IID environments without affecting privacy guarantees.

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